

OmniWarRoom AI

Autonomous Multi-Agent Swarm for Real-Time
Competitive Intelligence & Market Countermeasures

GTM Intelligence Track

Risk & Compliance Track

TECH STACK

-  **LangGraph 1.2.2**
Orchestration
-  **Bright Data MCP + REST**
Web Intelligence
-  **Groq LLaMA 3.1-8B**
Scout LLM
-  **Gemini 2.0 Flash**
Analyst LLM
-  **Groq LLaMA 3.3-70B**
Tactician LLM
-  **Cognee 1.1.0 + LanceDB**
Memory
-  **Pydantic v2 + Streamlit**
Schema & UI

The Problem & Our Answer

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THREE BLIND SPOTS

Your Competitors Are Invisible

- Competitors reprice overnight - you find out Monday morning.
- DataDome & Akamai block every classic scraper with a 403.
- Manual analysis takes hours, the response window is gone.

GTM Intelligence Track + Risk & Compliance Track

OMNIWARROOM ANSWER

Three Forces Working For You

- Autonomous worker loop scans all targets 24/7, never stops.
- Bright Data Web Unlocker bypasses all bot shields natively.
- Threat scored, plan written, alert sent - under 4 minutes.

INTERVAL = 300s | THRESHOLD = 3/5

Technical Stack

LAYER	TECHNOLOGY	ROLE IN PRODUCTION
 Orchestration	LangGraph 1.2.2	Async StateGraph, conditional routing scout->analyst->tactician->persistence
 Web Intelligence	Bright Data MCP + REST	Web Unlocker + SERP API - bypasses DataDome, Akamai, Cloudflare
 Scout LLM	Groq llama-3.1-8b-instant	Lightweight tool calling for MCP data collection, conserves TPM quota
 Analyst LLM	Gemini 2.0 Flash (primary)	Structured output via Pydantic MarketAnalysisOutput, cascade to Qwen3-32B
 Tactician LLM	Groq llama-3.3-70b-versatile	Counter-strategy plans (3-5 actions, ROI-framed) when threat >= 3
 Memory	Cognee 1.1.0 + LanceDB	Persistent knowledge graph (NetworkX) indexed via cognee.add / cognify()

Bright Data: The Intelligence Layer

WEB UNLOCKER

Bypass Every Bot Shield

- DataDome, Akamai & Cloudflare blocked on every classic scraper.
- Residential IP rotation + browser fingerprinting per request.
- Scout agent gets clean HTML - zero 403s at production scale.
- Raw output capped at MAX_RAW_CHARS = 3,500 chars (~875 tokens).

POWERS SCOUT NODE

SERP API

Google Search Intelligence

- Structured organic results + People Also Ask per query.
- Cross-referenced with Web Unlocker data by Analyst node.
- Parsed JSON returned - no HTML parsing, no retry logic.
- Output capped at MAX_SERP_CHARS = 1,500 chars (~375 tokens).

POWERS ANALYST NODE

MCP SERVER

LLM-Native Tool Calling

- streamable_http transport via MultiServerMCPClient.
- LLM binds tools dynamically - no hardcoded scraping logic.
- One endpoint replaces 10+ custom scraping integrations.
- Hackathon track: Bright Data AI Agents & Web Data.

POWERS TOOL DISPATCH

The Multi-Agent Swarm

SCOUT

llama-3.1-8b-instant

- Binds Bright Data MCP tools dynamically via LangGraph.
- LLM decides what to fetch - no hardcoded scraping.
- Web Unlocker: residential IP + browser fingerprint.
- Output capped: MAX_RAW_CHARS = 3,500 chars.

MAX_RAW = 3,500 chars

ANALYST

Gemini 2.0 Flash

- Scores threat 1-5 via Pydantic MarketAnalysisOutput.
- Cross-refs Web Unlocker + SERP + Cognee memory.
- Auto-cascade: Qwen3-32B -> Qwen2.5-7B on 429.
- Conditional: threat >= 3 -> Tactician node.

THRESHOLD = 3 / 5

TACTICIAN

llama-3.3-70b-versatile

- Fires only when threat_level >= THREAT_THRESHOLD.
- Generates 3-5 timed, ROI-framed counter-actions.
- Triggers email + Triggerware + pyttx3 in parallel.
- Fallback: llm_fallback on HTTP 429 rate limit.

3-5 ACTIONS / CYCLE

PERSISTENCE

Cognee 1.1.0

- Saves cycle to market_history.json (NDJSON).
- cognee.add() queues data to LanceDB + NetworkX.
- _bg_cognify() runs fire-and-forget (asyncio task).
- Analyst queries this graph in the next cycle.

MAX JOURNAL = 300 lines

The Autonomous Worker Engine

Process Isolation

worker.py is a standalone subprocess, fully decoupled from Streamlit ui/app.py.

True Parallelism

asyncio.gather() scans all competitor URLs simultaneously in each session cycle.

Atomic State Writes

Brain state written via `os.replace()` on a `.tmp` file - prevents race conditions.

Full Audit Trail

All events appended to `agent_journal.jsonl` - NDJSON, max 300 lines, auto-rotating.

Multi-Channel Dispatch

On critical threat: HTML email (SMTP auto-detect) + Triggerware webhook + pyttsx3 TTS.

Streamlit Interface

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CONTROL PLANE

Live Operations

- SVG brain polls worker_brain_state.json every 1.5s via native JS.
- Voice command: Speechmatics REST transcribes WebM audio to URL list.
- One-click Deploy Swarm with live execution trace in the terminal panel.
- STRATEGIC ALERT card expands / collapses via postMessage iframe API.

DATA PLANE

Market Intelligence

- KPI row: total scans, critical count, avg threat, latest alert host.
- Plotly time-series chart per host with green / amber / red zones.
- History feed: expandable cards per critical scan with signals + plan.
- Worker tab: phase matrix (Rich table), live journal, GTM alert feed.

Human-in-the-Loop

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ALERT SYSTEM

No Autonomous Action

- AI scores and plans - never prices, spends, or messages autonomously.
- _SMTP_MAP: 14 email domains auto-mapped (Gmail, Outlook, Yahoo, etc.).
- HTML email: threat gauge, signals list, numbered action plan, CTAs.
- Subject: [ACTION REQUIRED] Threat X/5 (LABEL) on competitor-host.

DATA SCHEMA

Pydantic Enforcement

competitor_name	str
------------------------	-----

Auto-identified from scraped page content

threat_level	int
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Normalised score 1-5 (threshold = 3)

signals_detected	list
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Category + description + severity per signal

confidence_score	float
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LLM certainty score from 0.0 to 1.0

action_plan	str
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Generated by Tactician - 3-5 timed actions

What Makes OmniWarRoom AI Different

Sees Through Every Wall

Bright Data Web Unlocker permanently removes DataDome, Akamai, Cloudflare. No 403s, no blind spots, clean HTML at production scale.

Production-Grade Swarm

Async LangGraph, asyncio.gather parallel scans, os.replace atomic writes, LLM cascade failover, Cognee knowledge graph. Built to run unsupervised.

4-Minute Closed Loop

Scout + Analyst + Tactician + Persistence. Raw web data becomes a qualified threat score and a ready counter-strategy in under 4 minutes, 24/7.

Human-in-the-Loop

AI analyses and recommends. Humans approve. Full audit trail in agent_journal.jsonl. Branded HTML alert with Approve / Reject CTA buttons.